

```
1 // Suppose we have two poses and velocities with timestamps.
2 Key p0 = Symbol('x', 0);
3 Key v0 = Symbol('v', 0);
4 double t0 = 0.0;
5 // Defined similarly for p1, v1, t1.
6
7 // Define states on (p0, v0) and (p1, v1) as timestamped pose-vel pairs.
8 StateData state0(p0, v0, t0);
9 StateData state1(p1, v1, t1);
10
11 // Suppose that p0, p1 are 2D points.
12 // Define a Power Spectral Density matrix of appropriate dimensions.
13 auto Q_point2 = Vector4::Ones();
14
15 // Define a WNOA factor for Point2.
16 WnoaMotionFactor<Point2> factor_P2(state0, state1, Q_point2);
17
18 // If p0, p1 are SE(3) poses instead, we would define the factor as
19 auto Q_se3 = Vector6::Ones();
20 WnoaMotionFactor<Pose3> factor_SE3(state0, state1, Q_se3);
```